

Lab 3: Component Modelling & Architectural Pattern Selection

Topic: Self-Service Coffee Kiosk System Architecture

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We chose [Layered Architecture] for the Self-Service Coffee Kiosk System

1. Architectural Choice & Scenario Justification

The Self-Service Coffee Kiosk operates as a dedicated on-premise embedded installation. A **Layered Architecture** organizes the system into Presentation, Business Logic, Peripheral Drivers, and Data Storage tiers:

- **Scenario Reason 1 (Hardware & Local Peripheral Control):** The kiosk directly manages physical hardware—a touchscreen display, card reader, and thermal receipt printer. Microservices introduce inter-process networking overhead, serialization latency, and deployment complexity that offer no benefit on a single localized terminal. Layered architecture provides clean separation of concerns while executing within a single reliable runtime.
- **Scenario Reason 2 (Component Modularity & Maintainability):** Encapsulating peripheral interaction inside the Printer Controller and payment handling in Payment Service ensures that updates to device drivers, receipt layouts, or terminal APIs do not destabilize the core ordering workflow or menu browsing.

2. Security Advantage

- **Privilege Separation & PCI-DSS Compliance:** Placing payment logic exclusively inside the Payment Service component behind a strict `IPaymentProcessing` interface enforces least-privilege access.
- The customer-facing Touchscreen UI has no direct access to cardholder data buffers or hardware pin-pad drivers, drastically minimizing the attack surface against memory injection or UI hijacking on the public kiosk terminal.

3. Performance Benefit

- **Zero Network Latency for High-Throughput Rush Hours:** A layered, monolithic deployment on local kiosk hardware enables in-process memory calls between Touchscreen UI, Order Manager, and Menu & Pricing Database.
- By avoiding remote HTTP/REST network hops for drink customization and price lookups, the kiosk maintains sub-second interaction speeds to rapidly process queues during peak café traffic.

